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Article *in* Power Systems, IEEE Transactions on · June 2017

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An Energy Sharing Model with Price-based Demand Response for Microgrids of Peer-to-Peer Prosumers

Nian Liu, *Member, IEEE*, Xinghuo Yu, *Fellow, IEEE*, Cheng Wang, Chaojie Li, Li Ma, Jinyong Lei

Abstract— According to the feed-in tariff for encouraging local consumption of Photovoltaic (PV) energy, the energy sharing among neighboring PV prosumers in the microgrid could be more economical than the independent operation of prosumers. For microgrids of peer-to-peer (P2P) PV prosumers, an energy sharing model with price-based demand response (DR) is proposed. First, a dynamical internal pricing model is formulated for the operation of energy sharing zone, which is defined based on the supply and demand ratio (SDR) of shared PV energy. Moreover, considering the energy consumption flexibility of prosumers, an equivalent cost model is designed in terms of economic cost and users' willingness. As the internal prices are coupled with SDR in the microgrid, the algorithm and implementation method for solving the model is designed on a distributed iterative way. Finally, through a practical case study, the effectiveness of the method is verified in terms of saving PV prosumers' costs and improving the sharing of the PV energy.

***Index Terms*— Smart grid, pricing, demand response, energy sharing.**

NOMENCLATURE

B_m	Upper bound of PV energy feed in the utility grid.
H	Total number of time slots in the operational time horizon.
n	Total number of PV prosumers participant in the energy sharing zone.
NP_i^h	Energy of net power of PV prosumer *i* in time slot *h*.
$NP_{i,\omega}^h$	Energy of net power of PV prosumer *i* in time slot *h* and scenario ω.
$\boldsymbol{P_i}$	PV energy of prosumer *i* in the operational time horizon.
P_i^h	Produced PV energy of prosumer *i* in time slot h.
$\boldsymbol{Pr}$	Internal price set in the operational time horizon.
Pr^h	Internal price in time slot *h*.
Pr_ω^h	Internal price in time slot *h* and scenario ω.
S	Income of the ESP in an operation day.
$\boldsymbol{TP_i}$	Energy of original power consumption of PV prosumer *i* in the operational time horizon.
TP_i^h	Energy of the original power consumption of PV prosumer *i* in time slot h.
TSP^h	Energy of total selling power in time slot *h*.
TSP_ω^h	Energy of total selling power in time slot *h* and scenario ω.
TBP^h	Energy of total buying power in time slot *h*.
TBP_ω^h	Energy of total buying power in time slot *h* and scenario ω.
TNP^h	Energy of total net power of the energy sharing zone in time slot *h*.
SDR^h	*SDR* of the energy sharing zone in time slot *h*.
SDR_ω^h	*SDR* of the energy sharing zone in time slot *h* and scenario ω.
Pr_{buy}	Internal buying price of the energy sharing zone.
Pr_{sell}	Internal selling price of the energy sharing zone.
$\boldsymbol{x_i}$	Adjusted power consumption profile of prosumer *i* in the operational time horizon.
x_i^h	Adjusted power consumption of PV prosumer *i* in time slot *h*.
inc_i	Equivalent cost of inconvenience of PV prosumer i.
$C_i^h(\cdot)$	Cost function of PV prosumer *i* in time slot *h*
$C_{i,\omega}^h(\cdot)$	Cost function of PV prosumer *i* in time slot *h* and scenario ω.
α_i	Sensitivity coefficient of PV prosumer *i*.
β	Percentage of service fee by shared energy.
λ^{buy}	Unit price of buying energy from the utility grid.
λ^{sell}	Unit price of selling energy to the utility grid.
τ	Current time slot.
$\delta_{i,\boldsymbol{\omega}}^h$	Related error (%) of forecasted value of PV prosumer *i* in time slot *h* and scenario ω.

I. Introduction

In recent years, Photovoltaic (PV) systems have grown rapidly worldwide. Influenced by solar radiation and temperature, the output power of the PV system has the characteristics of randomness, volatility, and intermittency. Considering the power balance requirement, the large-scale connection of PV systems will have significant influence on the operation of the electric power system. Thus, at the present stage, many countries' energy policies aim to encourage the self-consumption of PV energy from the customer's perspective [1] [2]. The microgrid is regarded as the preferred way to promote the integration of distributed PV systems. The power of PV systems could be maximally consumed within the microgrid which reduces the impact on the utility grid [3].

This work was supported by the National High-tech R&D Program of China (863 Program) (No. 2014AA052001).

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Through the relationship of parties participating in energy supply and demand, the microgrid can be divided into three different categories. The first is one party with self-supply and self-consumption. For this type of microgrid, the objective of energy management is to minimize the operation cost [4] [5] [6]. The second is two parties in the hierarchy, one for microgrid operator (or distributed generation operator), and the other for electricity users [7] [8] [9]. Usually, the operator decides the electricity price based on the demand and the marginal cost of operation to maximize the profit. The users schedule the electrical appliance based on the price to minimize the cost. The third is multiple parties, such as a number of PV prosumers [10] [11] [12]. Every prosumer can either act as an energy supplier or consumer, and they have equal privilege and are equipotent participants in the microgrid operation. Referring to the Peer-to-Peer (P2P) networking defined in Computer Science, the third type microgrid can be named as microgrid of P2P prosumers.

For the microgrid of P2P prosumers, the trading of surplus PV energy among prosumers is an important issue for the economic operation [13]. In order to deal with the internal energy trading problem of microgrids, one of the proposed solutions is to utilize auction models and multi-agent (MA) theory to treate the microgrid as a small energy market. The energy exchange between the production units and the local loads of the microgrid is formulated as a symmetrical assignment problem [10]. The MA system and a classical distributed algorithm are used to solve the problem. A real-time MA-based platform for microgrid market operation is presented [11]. A competitive game-theory reverse auction model is investigated to schedule the distributed energy resource commitment with an hour-ahead market approach. Similarly, a simulation platform based on JADE and MATLAB/Simulink is presented in [12]. Actions of distributed generators and price-sensitive loads participating in the microgrid energy market are simulated. All these methods are based on the electricity market of microgrid, which requires a complete auction and bidding mechanism among the prosumers. However, the problem is that some countries don't have electricity markets on the demand side, and their energy policies only regulate the feed-in tariff of PV energy.

On the other hand, there are also researches that focus on a simple way for energy sharing or trading among neighboring prosumers. In [14], an energy sharing approach is developed to determine the energy profile matching among the prosumers with minimized cost. In [15], an autonomous cooperative energy trading is designed based on policies and protocols for sharing and matching of energy schedules. However, the problem of these methods is the lack of a pricing mechanism. Without a price, it is difficult to facilitate the energy sharing with the demand response (DR). Meanwhile, the pricing of electricity is one of the main research areas in Smart Grid. One category is for the pricing problem between retailers and end users. For instance, a bi-level stochastic programming was proposed for the retailer to decide the prices that are offered to its potential end users [16]. In [17], a Stackelberg game model was proposed to optimize the day-ahead hourly prices by means of the interactions between the retailer and customers. By using Walrasian auctions, the design and evaluation of a market clearing scheme for the exchange of DR was proposed [18]. The DR is regarded as a virtual resource that can be traded between DR buyers (TSO, distributors, retails) and DR sellers (aggregators, customers). Moreover, there is another category of pricing problem on the demand side, with the utilization of renewable energy resources (RESs). For instance, a DR model was built between the virtual power plant (VPP) and consumers, and the pricing was designed based on a fixed basic price plus an additional profit distribution [19]. In [20], a real-time direct trading between small-scale suppliers and end users was modelled as a coalitional game. The Shapley value is used as a means of deriving the trading prices.

In this paper, the focus is on the energy sharing problem inside the microgrid of P2P PV prosumers. Compared with aforementioned studies, there are mainly two differences. First, during different time periods, prosumers may change their behaviors to act as sellers or buyers depending on circumstance and net power profile. Moreover, as the prosumers have equal privilege and are equipotent participants in the microgrid, the energy sharing prices should be co-decided by all the prosumers. This is a special kind of pricing problem and has not attracted much attention in the literature of Smart Grid, to the best of our knowledge. To this end, this paper proposes an energy sharing model with price-based DR for microgrids of P2P prosumers. The main contributions are as follows.

1) An energy sharing structure is proposed to integrate the autonomous PV prosumers into an energy sharing zone, and a virtual entity named energy sharing provider (ESP) is defined to coordinate the sharing activities.

2) A dynamical internal pricing model is proposed for the operation of energy sharing zone, which is defined based on the supply and demand ratio (SDR) of shared PV energy, and the feed-in tariff is considered as the price bound.

3) As the internal prices are adjusted with the change of SDR, the autonomous demand response of participants is integrated into the energy sharing, and the implementation method is also designed.

The rest of this paper is organized as follows. Section II describes the energy sharing structure in microgrids. Section III introduces the basic model of PV prosumers and microgrids. In Section IV, the energy sharing model is formulated, including the internal pricing model and the cost model of PV prosumers. Section V outlines the implementation and algorithms for the model. Case studies and analysis of results are presented in Section VI. Conclusions are given in Section VII.

II. Energy Sharing Structure of PV Prosumers

A. *Typical metering configuration of PV prosumers*

According to feed-in tariffs of many countries, the typical metering configuration of PV prosumers is shown in Fig. 1. There are generally two separate meters configured, one is used for metering of PV generation, the other for bi-directional metering of import and export electricity of the PV prosumer. Correspondingly, there are three types of prices designed for PV prosumers, namely generation price, import price, and export price. The generation price is set for total generated energy (kWh) of the PV system, no matter the electricity is self-consumed or exports to the utility grid.

Sometimes, the generation price is implemented in the form of subsidy. In order to encourage self-consumption, export price is usually lower than import price. For example, according to China's feed-in tariff, the generation, export, and import prices could be respectively 0.42, 0.4, and 0.5-1.2 CNY/kWh, in which the generation price is implemented in the form of government subsidy. For this type of feed-in tariff, the self-consumption of PV energy is greatly encouraged. This will possibly push the users to reschedule the electricity consumption based on the PV generation curves.

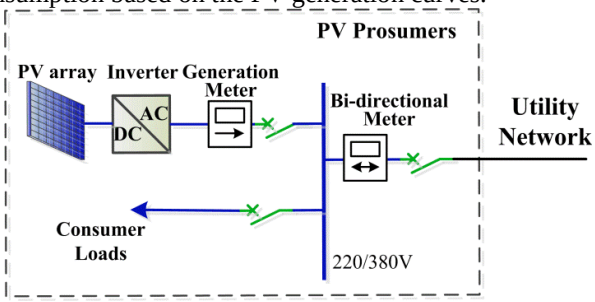

Fig. 1 Typical metering configuration of PV prosumers.

B. *Energy sharing structure*

The energy sharing structure of the microgrid is shown in Fig. 2. Each PV prosumer is comprised of PV system, load, user energy management system (U-EMS), and smart meters etc. Based on this structure, the PV prosumers can participate in the energy sharing zone, and their extra PV energy can be shared among prosumers, instead of directly trading with the utility grid. Although power curves of PV systems have high similarity due to almost the same solar radiation inside a microgrid, the net power varies widely among PV prosumers, which have different load profiles and PV capacities. Thus, the PV power is possible to be shared inside a microgrid.

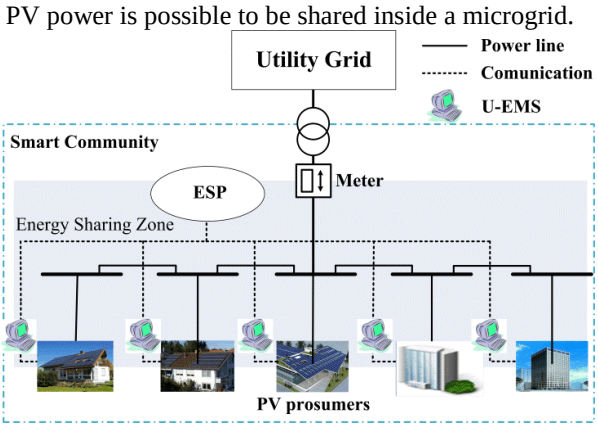

Fig. 2. Energy sharing structure of the microgrid.

In order to facilitate the organization of energy sharing, a coordinator is required. Here, we introduce a new entity named Energy Sharing Provider (ESP). Generally, the ESP may belong to the microgrid operator or an independent third party company that focuses on the business mode of energy sharing, which provides energy sharing service for all participated PV prosumers. The ESP acts as an agent for prosumers and is responsible for assuring the power balance and the payment balance in the energy sharing zone. In this regard, the ESP is in charge of purchasing electricity from the utility grid, or from the PV prosumers with extra net power export, then selling electricity to other prosumers or the utility grid. In the energy sharing zone, self-consumption of PV energy is still the first choice for PV prosumers. If PV energy can not satisfy the load demand, the PV prosumer would purchase energy agented by the ESP. If the demand of the PV prosumer is less than the production of PV source, the redundant PV power will be shared with other PV prosumers or sold to the utility grid via the ESP. In order to keep the fairness of energy sharing, the ESP should not have an optimal objective with any tendency, but it still needs to charge for service fee with the prosumers, as to cover the power losses and other possible expenditures.

Although the feed-in-tariff can provide basic motivation for the energy sharing, we still need an appropriate internal pricing mechanism. The reasons mainly include two aspects: i) the electricity prices are a basic requirement for billing among the prosumers; ii) the dynamic pricing-based DR is conducive to improve the utilization of PV energy inside the microgrid.

To assist the energy sharing and DR, each prosumer is assumed to have a U-EMS that can be used to collect data of PV source and load, and receive the price information from the ESP. In addition, the U-EMS should in charge of optimal scheduling for shiftable appliances. During the operation, a forecasting function is integrated into the U-EMS to provide power predictions of the PV systems. Moreover, the bi-directional communications between the ESP and PV prosumers are required for the energy sharing activities. Generally, these requirements are the basic configuration of demand side energy management in Smart Grid [21].

III. Energy Sharing Model

A. *Basic model*

Generally, all PV prosumers are composed of different loads and PV systems. Before participating in an energy sharing zone, the original power consumption pattern of PV prosumer *i* during the operation time period is defined as:

$$\boldsymbol{TP_i} \triangleq [\mathrm{TP}_i^1, \mathrm{TP}_i^2, \dots, \mathrm{TP}_i^H], \quad i \in [1,2,\dots,n] \tag{1}$$

where n is the total number of PV prosumers in the microgrid, and H is the number of time slots over the operation time period, which is 24 hours in this study.

During the operation, the PV system usually uses the maximum power point tracking (MPPT) control to maximize the power output. As the power of PV system changes with solar intensity and environment temperature, the predicted active power of PV prosumer *i* is defined as:

$$\boldsymbol{P_i} \triangleq [P_i^1, P_i^2, \dots, P_i^H] \tag{2}$$

For PV prosumer *i*, its net power at time slot *h* can be defined as:

$$\mathrm{NP}_i^h = TP_i^h - P_i^h, h \in [1,2,\dots,H] \tag{3}$$

As the loads and PV output of prosumers are different from each other, they would act on selling role or buying role at a certain time slot based on the net power. Thus, the Total Selling Power (TSP) and Total Buying Power (TBP) at time slot *h* can be calculated as:

$$TSP^h = -\sum_{i=1}^{n} NP_i^h, \quad NP_i^h < 0 \tag{4}$$

$$TBP^h = \sum_{i=1}^{n} NP_i^h, \quad NP_i^h \geq 0 \tag{5}$$

Furthermore, the total net power of the energy sharing zone at time slot *h* (denoted as TNP^h) can be calculated as:

$$TNP^h = TBP^h - TSP^h = \sum_{i=1}^{n} NP_i^h \quad (6)$$

B. Internal price model

Before designing energy sharing prices, we mainly consider three basic principles: i) the internal prices should be bounded between the feed-in tariff and electricity prices of utility grid; ii) as the basic principle of economics, the relation between price and SDR is inverse-proportional; iii) the economic balance should always be guaranteed for the energy sharing zone.

Supply and demand are fundamental concepts of economics. In the energy sharing zone, demand refers to how much (quantity) of power energy is desired by buyers, and supply represents how much the sellers can offer. As there is no obvious leader in the energy sharing zone, the reflection of supply and demand is the foundation to decide internal prices, which can be co-decided by all the prosumers. Specifically, the relation of supply and demand in the energy sharing zone can be represented by the total selling power TSP^h and total buying power TBP^h. Thus, the SDR of PV energy at time slot *h* is defined as:

$$SDR^h = \frac{TSP^h}{TBP^h} \quad (7)$$

Considering the fluctuation of PV power and users' load, the *SDR* is also not constant at different time slots. In order to reflect the fluctuation, the Internal Price Set (IPS) is defined based on selling and buying prices at different time slots.

$$\mathbf{Pr} \triangleq \{Pr_{buy}^1, Pr_{buy}^2, \dots, Pr_{buy}^H; Pr_{sell}^1, Pr_{sell}^2, \dots, Pr_{sell}^H\} \quad (8)$$

As shown in Fig. 3, the ESP acts as an agent for all the PV prosumers. The ESP purchases power from PV prosumers and utility grid at the unit price of Pr_{buy} and λ^{buy}, and sells power to PV prosumers and utility grid at the unit price of Pr_{sell} and λ^{sell}, respectively.

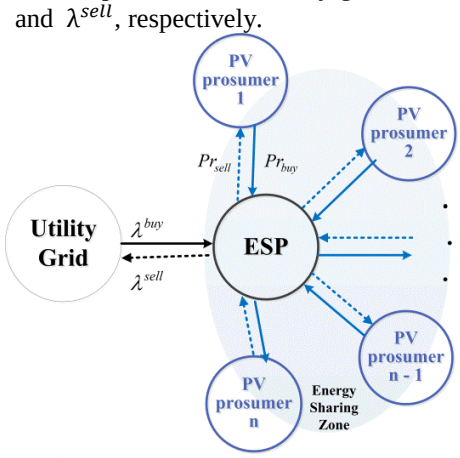

Fig. 3. Prices of energy exchanges among the utility grid, ESP and prosumers.

Generally, the relationship between price and SDR is inverse-proportional [22], and thus, we can use this principle to formulate the pricing model. Firstly, with the constraints of feed-in-tariff, the internal selling price is defined as a piecewise function of SDR:

$$Pr_{sell}^h = f(SDR^h) = \begin{cases} \frac{\lambda^{sell} \cdot \lambda^{buy}}{(\lambda^{buy} - \lambda^{sell}) \cdot SDR^h + \lambda^{sell}}, & 0 \le SDR^h \le 1 \\ \lambda^{sell}, & SDR^h > 1 \end{cases} \quad (9)$$

When *SDR*=0, which means no PV prosumer sells power, the required energy is bought from the utility grid at the price of λ^{buy}; When $SDR \geq 1$, which means the power is oversupply, the redundant PV power has to be sold to utility grid at λ^{sell}. Only when $0 < SDR^h < 1$, the selling price would dynamically change between λ^{sell} and λ^{buy}.

Secondly, the internal buying price is defined as an average value considering the cost of internal selling energy, utility grid's charge and the ESP's charge:

$$Pr_{buy}^h = f(SDR^h) = \begin{cases} Pr_{sell}^h \cdot SDR^h + \lambda^{buy}(1 - SDR^h), & 0 \le SDR^h \le 1 \\ \lambda^{sell}, & SDR > 1 \end{cases} \quad (10)$$

When $0 < SDR^h < 1$, the total power demand exceeds the total power supply of PV systems in the energy sharing zone, the lacking power is bought from the utility grid at the price of λ^{buy}.

The reasoning detail of the internal price model is provide in Appendix I. Moreover, the relationship between the internal price and *SDR* is shown in Fig. 4, in which Pr_{sell} and Pr_{buy} decline with the increasing of *SDR* on the interval of [0, 1]. Generally, as the PV power is an uncontrolled value for maximizing utilization, the SDR is only adjustable by the load scheduling of PV prosumers. From the rationality viewpoint of the prosumers, if the SDR is small, the buyers would like to reduce the power consumption for a lower buying price, and the sellers would also like to reduce the power consumption to sell more energy. Thus, SDR could be increased by reducing power consumption. In contrast, if the SDR is large, the sellers would like to increase the power consumption for a higher selling price, and the buyers would also like to increase the power consumption for more energy. Finally, the SDR could be decreased by the increasing power consumption. We assume that every prosumer has a certain proportion of shiftable loads, that is, the prosumers have the ability to change the SDR, and consequently affect the final internal prices of the energy sharing zone. Hence, the internal prices can logically draw the activities of prosumers by DR.

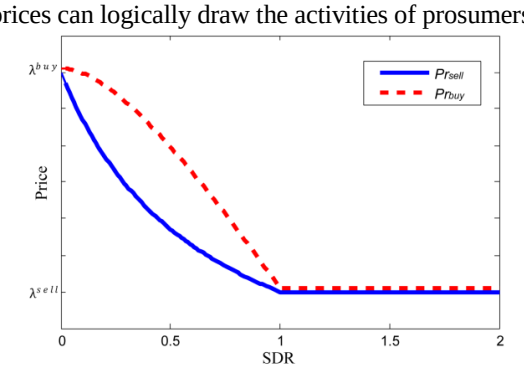

Fig.4. Relationship between the internal prices and SDR.

C. Cost model of the PV prosumer

Here the PV prosumer is a new emerging types of entity who consumes energy and also produce energy, i.e., smart building, factory, house, etc. In macro grid market, the utility block model [23] and elastic load demand model [16] are generally used to formulate the optimal pricing model. For the different level of load quantity, the models cannot be directly applied for prosumers.

All PV prosumers participate in the DR of energy sharing are assumed to have a certain proportion of shiftable load. Due to the price incentive, the PV prosumers may decide to

change the profiles of power consumption, which makes their $\boldsymbol{TP}$ deviate from the original value. The adjusted power consumption profile can be denoted as:

$$\mathbf{x}_{\boldsymbol{i}} \triangleq [x_i^1, x_i^2, \dots, x_i^H] \tag{11}$$

where x_i^h is adjusted power of the *i*-th PV prosumer at time slot *h*.

For PV prosumer *i*, its adjusted net power at time slot *h* can be updated as:

$$NP_i^h = x_i^h - P_i^h, h \in [1,2,\dots,H] \tag{12}$$

Recently, the users' willingness to shift load have been considered as an important factor in the energy management of demand side, since their sensitiveness toward load shifting are different [24]. As for prosumers, if they use the shiftable appliances without any concern of the DR incentives, they cannot feel any inconvenience, and thus the original load profile will not change. Otherwise, if they decide to adjust the shiftable load in response to the prices, the actions will apparently affect their convenience that the appliances would be delayed or shifted to an earlier period from the original period. Essentially, the inconvenience caused by load shifting is more accurate to be measured by the number and time changes of the shiftable appliances. As discussed in Appendix II, with respect to power consumption, the equivalent cost of inconvenience can be defined as:

$$inc_{\boldsymbol{i}} = \alpha_i \sum_{h=1}^{H} (x_i^h - TP_i^h)^2 \tag{13}$$

where inc_i is the equivalent inconvenience cost of PV prosumer *i*; α_i is the sensitivity coefficient of PV prosumer *i*, which is used to quantify the prosumers' willingness towards the load shifting. For the selection of α_i ($\alpha_i > 0$), the larger value indicates that the prosumer is more sensitive to the inconvenience caused by the load shifting, and has less willingness to adjust the load usage.

Integrating the inconvenience cost with the electricity cost or income of the PV prosumer, the cost function for optimal operation of PV prosumers can be defined as:

$$C_i^h(x_i^h) = Pr_i^h(x_i^h - P_i^h) + \alpha_i(x_i^h - TP_i^h)^2 \tag{14}$$

$$C_i = \sum_{h=1}^{H} C_i^h(x_i^h) \tag{15}$$

where $C_i^h(x_i^h)$ can be separated into two parts: cost for using electricity $Pr_i^h(x_i^h - P_i^h)$ and equivalent cost for adjusting flexible power $\alpha_i(x_i^h - TP_i^h)^2$. These two parts constitute the cost function in both economy and users' willingness. Pr_i^h is the price at time slot *h* for prosumer *i*, which can be selling price or buying price, and decided by the net power:

$$Pr_i^h = f(NP_i^h) = \begin{cases} Pr_{sell}^h, & NP_i^h < 0 \\ Pr_{buy}^h, & NP_i^h \geq 0 \end{cases} \tag{16}$$

As the PV prosumers can act as selling or buying roles, there are two conditions for the value of $Pr_i^h \cdot NP_i^h$:

1) Selling role: when $NP_i^h > 0$, the prosumer needs to purchase power at the cost of $Pr_i^h \cdot NP_i^h$ and then minimizing $C_i^h(x_i^h)$ is to minimize its cost.

2) Buying role: when $NP_i^h < 0$, the prosumer needs to sell power, and thus minimizing $C_i^h(x_i^h)$ is equal to the maximization of PV prosumer's profit.

Furthermore, there are three constraints for the energy scheduling:

$$\sum_{h=1}^{H} x_i^h = \sum_{h=1}^{H} TP_i^h \tag{17}$$

$$\min(\boldsymbol{TP_i}) \leq x_i^h \leq \max(\boldsymbol{TP_i}) \tag{18}$$

$$P_i^h - x_i^h \leq B_m \tag{19}$$

where (17) means the flexible load can be shifted to a different time slot, but the total energy requirement is assumed to be constant [24] [25]; (18) is used to assure the reality of electricity usage, which can't cut down users' baseload and exceed the upper limit of supply capacity; (19) means the feed-in power should be constrained when the utility grid prohibits extra generated power for grid security reasons.

As shown in Fig.5, we can see that equation (14) is a convex function. Firstly, the gradient of curves increases with α_i, which reflects that the prosumers are sensitive to the convenience. Secondly, the minimum values of $C_i^h(x_i^h)$ also increase with α_i, which means that the prosumers need to pay more for the enhancement of convenience level.

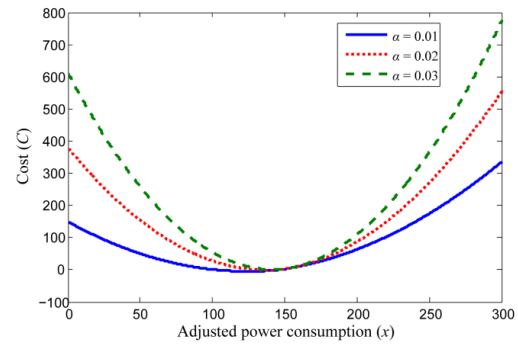

Fig. 5. Impact of α_i on the cost functions of PV prosumers. ($Pr_{sell} = 0.5498$, $Pr_{buy} = 0.7543$, $P_i^h = 147.24$, $TP_i^h = 152.7$.)

D. Income model of the ESP

As the agent of PV energy sharing, the ESP could charge the service fee from the participating prosumers, which is directly proportional to the amount of shared PV energy. The income of the ESP in an operation day is defined:

$$S = \beta \sum_{h=1}^{H} \min(TSP^h, TBP^h) \tag{20}$$

where β is the charged percentage of shared energy.

For this model, there are problems that need to be clarified: i) the distribution of service fee; ii) the value of β. First, we think it is fair for all the prosumers to pay for the service fees according to their contributions of shared energy. The service fee distribution model can be formulated as:

$$S_i^h = \frac{\beta \cdot |NP_i^h| \cdot \min(TSP^h, TBP^h)}{TSP^h + TBP^h} \tag{21}$$

Moreover, for the value of β, the service fee should cover the losses of sharing power and other possible expenditures for organizing the sharing activities. In fact, the ESP is a virtual entity that cannot produce any PV energy, and thus the profit should be limited for actual applications.

IV. Implementation

A. Bilevel model

The optimal problem of the energy sharing can be reformulated as a bi-level programming model, such that

Upper level: find a fixed point for each Pr_i^h

$$Pr_i^h = f(NP_i^h) = \begin{cases} Pr_{sell}^h, & NP_i^h < 0 \\ Pr_{buy}^h, & NP_i^h \geq 0 \end{cases} \tag{22}$$

where the nonlinear algebraic mappings for Pr_{sell}^h and Pr_{buy}^h are expressed in (7), (9) and (10).

Lower level:

$$\begin{cases} \min_x \mathrm{C}_i(\boldsymbol{x}_i, \boldsymbol{Pr}_i) = \\ \sum_{h=1}^{H}\left(Pr_i^h(x_i^h - P_i^h) + \alpha_i(x_i^h - TP_i^h)^2\right) \\ s.t. \quad \sum_{h=1}^{H} x_i^h = \sum_{h=1}^{H} TP_i^h \\ \min(\boldsymbol{TP_i}) \le x_i^h \le \max(\boldsymbol{TP_i}) \\ P_i^h - x_i^h \le B_m \end{cases} \tag{23}$$

Note that there is no coupled constraint between two prosumers. Different prosumers in the energy sharing zone are coupled by the pricing model. Thus the lower level programming can be modeled only with respect to prosumer i. All prosumers can solve their bi-level optimization problems locally. At the upper level, the goal is to find a fixed pricing strategy for each prosumer i at each time slot h. At the lower level, the objective is to minimize the cost or maximize the benefit of all the prosumers.

Generally, the bi-level optimal problem can be solved in centralized manner in form of equilibrium problem with equilibrium constraints (EPEC) [26]. However, in the upper level, our proposed model does not have an explicit objective function, which brings difficulty to directly formulate the KKT condition. Therefore, we have proposed an iterative algorithm to solve the problem.

Considering the forecasting error of PV power is always existed, the related error (%) of forecasted value of PV prosumer i in scenario ω is defined on a probability space (Ω, Γ, π) as:

$$\boldsymbol{\delta_{i,\omega}} \triangleq \left[\delta_{i,\omega}^1, \delta_{i,\omega}^2, \dots, \delta_{i,\omega}^H\right] \tag{24}$$

For the upper level price model, the decision has no direct relation with time horizon, the internal prices can be decided as day-ahead or hour-ahead (real-time). For the lower level problem, the original model is designed for day-ahead scheduling, but it can be decomposed to an hour-ahead model considering the uncertainty of forecasting error as:

$$\begin{cases} \min_x \mathrm{C}_i(\boldsymbol{x}_i, \boldsymbol{Pr}_i) = \mathrm{C}_i^\tau(x_i^\tau) + E\left(\sum_{h=\tau+1}^{H} \mathrm{C}_{i,\omega}^h(x_i^h)\right) \\ s.t. \quad \sum_{h=1}^{H} x_i^h = \sum_{h=1}^{H} TP_i^h \\ \min(\boldsymbol{TP_i}) \le x_i^h \le \max(\boldsymbol{TP_i}) \\ P_i^h(1 + \delta_{i,\omega}^h) - x_i^h \le B_m \end{cases} \tag{25}$$

where τ is the current time slot, and $\mathrm{E}[\cdot]$ is the expectation cost of prosumer i for all scenarios $\omega \in \Omega$.

Due to the stochastic nature of the vector $\boldsymbol{NP}_{i,\omega}$ in the function, the closed form of $E\left(\sum_{h=\tau+1}^{H} \mathrm{C}_{i,\omega}^h(x_i^h)\right)$ cannot be obtained while $Pr_{i,\boldsymbol{\omega}}^h\left(x_i^h - P_i^h(1 + \delta_{i,\boldsymbol{\omega}}^h)\right)$ is easily computable for given x_i^h, P_i^h and $\delta_{i,\boldsymbol{\omega}}^h$ in any scenario ω. To numerically obtain the cost function in (24), the sampling average approximation method (SAA) [27] [28] is applied for approximating the expectation cost as follows:

$$E\left(\sum_{h=\tau+1}^{H} \mathrm{C}_{i,\omega}^h(x_i^h)\right) = \frac{1}{M}\sum_{m=1}^{M}\sum_{h=1}^{H}\left(Pr_{i,\boldsymbol{\omega_m}}^h\left(x_i^h - P_i^h(1 + \delta_{i,\boldsymbol{\omega_m}}^h)\right) + \alpha_i \cdot \left(x_i^h - TP_i^h\right)^2\right) \tag{26}$$

B. Iterative algorithm

The interaction process for the energy sharing with DR is shown in Fig. 6. The overall process is implemented in a closed-loop iterative way. Optimal scheduling for each PV prosumer could be carried out according to the received internal price, considering the forecasting PV power, load consumption, and user's willingness. During the iterative optimization, the information exchange between the ESP and PV prosumers are listed in Table I.

TABLE I
INFORMATION EXCHANGE BETWEEN THE ESP AND PV PROSUMERS

Time Horizon		ESP	Prosumer i
Day-ahead	Send	$\mathbf{Pr}$	$\boldsymbol{NP}_i$
	Receive	$\boldsymbol{NP}_i, i \in [1,2,\dots,n]$	$\mathbf{Pr}$
Hour-ahead	Send	$Pr_{\boldsymbol{\omega_m}}^h$, $h \in [\tau, \tau+1, \dots, H]$	NP_i^h, $h \in [\tau, \tau+1, \dots, H]$
	Receive	NP_i^h, $h \in [\tau, \tau+1, \dots, H]$, $i \in [1,2,\dots,n]$	$Pr_{\boldsymbol{\omega_m}}^h$, $h \in [\tau, \tau+1, \dots, H]$

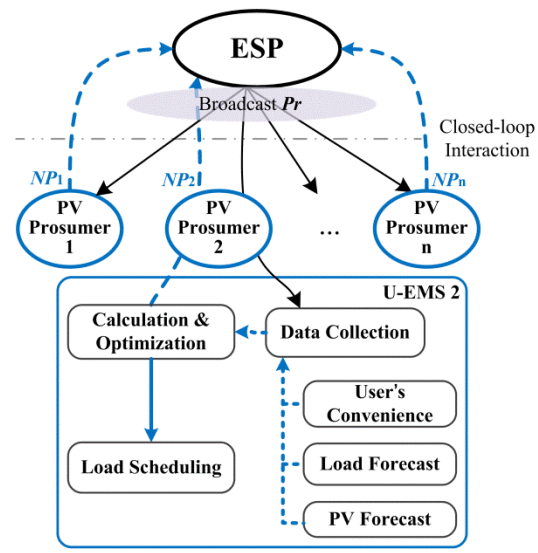

Fig. 6 Interaction between the ESP and PV prosumers.

For each step scheduling, PV prosumers' net power might be changed, and the new results will be sent to the ESP. Then the ESP recalculates and broadcasts the updated internal prices. After enough iterations, if all the PV prosumers can not adjust the scheduling anymore, the final internal prices for PV energy sharing is decided. Thus, the question is whether load and price pairs will finally converge to a fixed point in a DR-integrated energy sharing zone.

In the Appendix III, we have analyzed the existence of the fixed point for the proposed energy sharing model. In order to get a fixed point to implement the energy sharing in the microgrid, a distributed algorithm (as shown in Algorithm 1) is proposed based on the *Brouwer Fixed Point Theorem* [29] [30]. It is noted that the convergence may be affected by the DR participation level [31]. Therefore, the solutions of distributed algorithm may not converge to a fixed point under every scenario and it is up to the sensitivity coefficient α, which is used to indicate the willingness of participators. If the iterative process converges to a fixed point after a finite number of iterations, the energy sharing zone with internal price $\boldsymbol{Pr}$ converges and all PV prosumers can benefit from this feasible solution. Otherwise, it is divergent, the energy sharing model is unsolvable and the PV prosumers still can trade with the utility grid directly.

Algorithm 1

Initialize $\boldsymbol{TP_i}, \boldsymbol{P_i}, ,i = [1,2,\dots,n]$

for k=1 to K **do**

Collect $\boldsymbol{NP_i}$ from each U-EMS, randomly sample $\delta_{i,\omega}^h$ of M realizations to generate $\boldsymbol{NP_{i,\omega_m}}$;

Calculate *TSP* and *TBP*:

$$TSP_{\omega_m}^h = \sum_{i=1}^n NP_{i,\omega_m}^h, \quad NP_{i,\omega_m}^h < 0$$

$$TBP_{\omega_m}^h = \sum_{i=1}^n NP_{i,\omega_m}^h, \quad NP_{i,\omega_m}^h \geq 0$$

Calculate *SDR*: $SDR_{\omega_m}^h = \frac{TSP_{\omega_m}^h}{TBP_{\omega_m}^h}$

Central setting price: $Pr_{\omega_m}^h = f\left(SDR_{\omega_m}^h\right)$,

Local optimization at each U-EMS of PV prosumer:

$$\mathbf{x}_i = \underset{x}{\arg\min} \frac{1}{M}\sum_{m=1}^M \sum_{h=1}^H (Pr_{i,\omega_m}^h (x_i^h - P_i^h(1 + \delta_{i,\omega_m}^h)) + \alpha_i \cdot \left(x_i^h - TP_i^h\right)^2)$$

if $\|\boldsymbol{Pr} - \boldsymbol{Pr}_{old}\| \leq \varepsilon$ **then**

stop iteration

end if

end for

if k>K **then**

The energy sharing model is unsolvable, and then PV prosumers trade with utility grid directly.

else

The internal price is settled as $\boldsymbol{Pr}$, PV prosumers agree to share the surplus PV energy in the microgrid.

end if

V. Case Study

A. Basic data

Actual data of a demonstration project of China Southern Grid is used for the case study. In the system, there are residential buildings (RB), commercial buildings (CB) and office building (OB), as shown in Fig. 7. The parameters of the buildings are shown in TABLE II. All of the buildings have installed PVs with capacity ranging from 100 to 400 kW, and the PV output curves of typical day are shown on Fig. 8 (a). The daily load curves, daily net power curves of these prosumers are shown in Fig. 8 (b) and Fig.8 (c), respectively. The curves of total load and net load of the energy sharing zone are shown in Fig. 8 (d). PV generation forecasting errors are modelled by a Gaussian distribution, the pdf of errors can be found in [32].

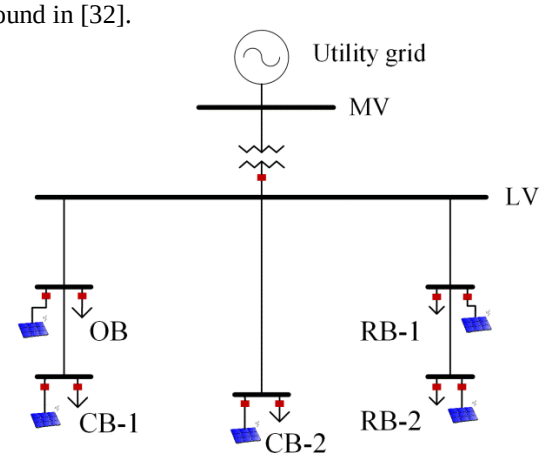

Fig. 7 Study case network of the microgrid.

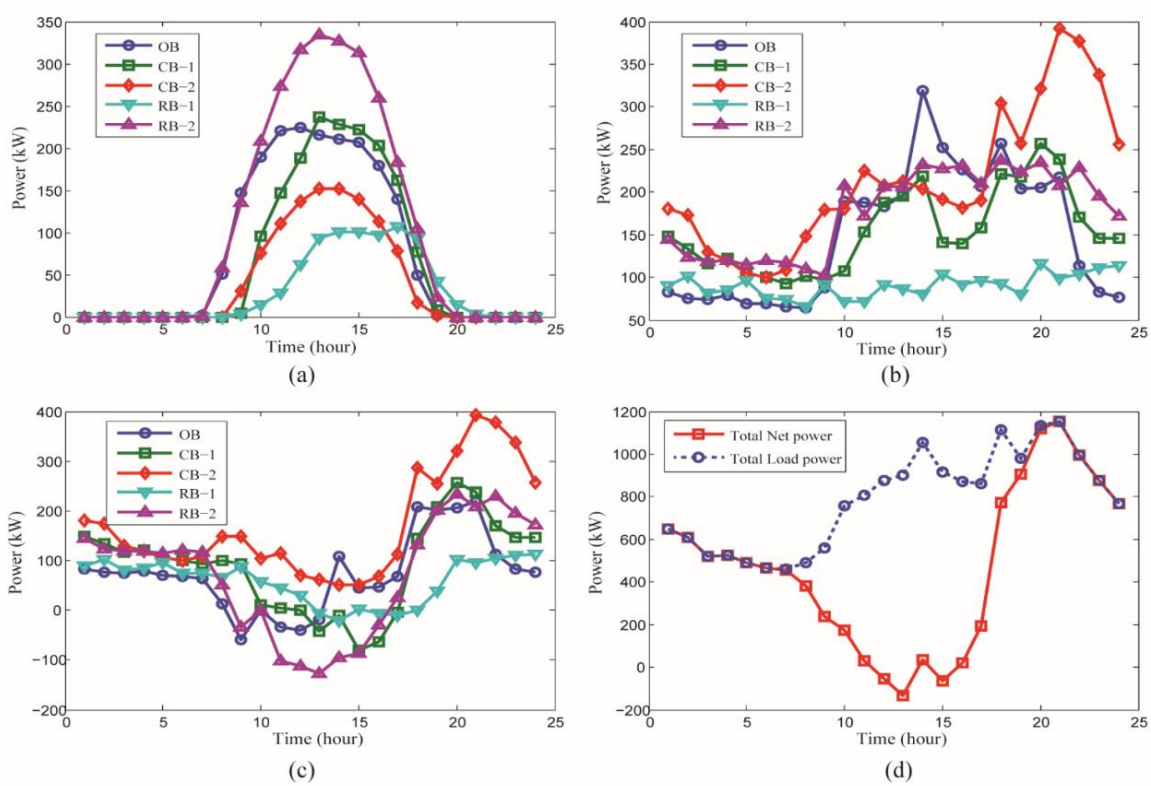

Fig. 8. Load/power curves of the typical day in summer: (a) prosumers' PV power curve; (b) prosumers' load curve; (c) prosumers' net power curve; (d) total load curve and net power curve.

TABLE II
PARAMETERS OF PV PROSUMERS IN THE MICROGRID

Name of the PV prosumers	Capacity of PV(kWp)	The maximum load in the typical day(kW)
OB	250	319.5
CB-1	250	256.8
CB-2	150	392.4
RB-1	100	116.1
RB-2	400	236.8

As shown in Fig. 8(d), the peak of total load and net power in the energy sharing zone is about 1200 kW at 20:00. There is a large quantity of PV power used by PV prosumers around 13:00. As PV prosumers have different load characters and power outputs, they possibly share the surplus PV energy inside the microgrid.

Based on the feed-in tariff of China, the prices λ^{sell} and λ^{buy} are set as 0.4 and 1.0 (CNY/kWh). Other parameters α, β, ε, K and M are set as 0.01, 0.12, 1 E-4, 100, and 30, respectively.

B. *Results of internal prices*

By using the model and algorithm, the deviations of selling and buying prices between two iterations are shown in Fig. 9, respectively. The final internal prices of different time slots are shown in Fig. 10, and the corresponding total net power is shown in Fig. 11.

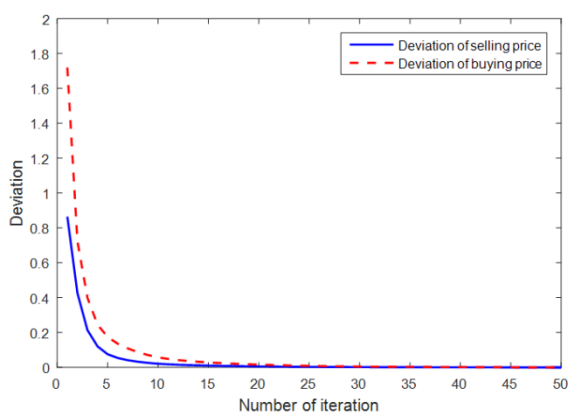

Fig. 9. Deviations of prices between two iterations.

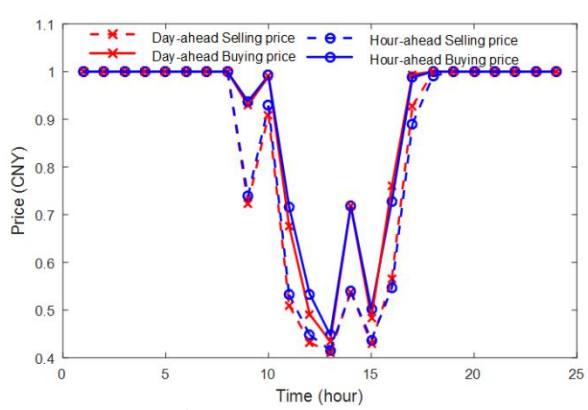

Fig. 10 Internal prices of the operational day.

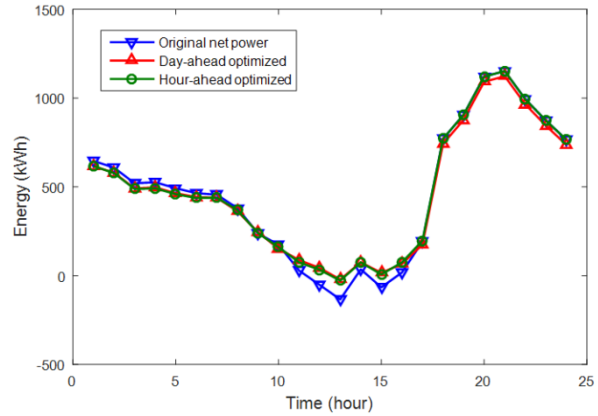

Fig. 11. Curves of net power of the operational day.

The difference between two iterations is convergent after dozens of iterations. When the internal prices are finally accepted, PV prosumers' load consumptions will not change because they have got the minimum cost on their own.

According to (7) – (10), the internal prices of every time slot are decided by the SDR. From Fig. 4, it is obvious that both selling and buying prices have the same changing tendency with the increasing of SDR. As shown in Fig. 10, the PV power is shared among PV prosumers from 7:00 to 17:00. And the finally decided internal prices always conform to the relationship between selling and buying prices showed in Fig. 4. Moreover, in Figs. 10 and 11, the results show that the hour-ahead optimization model has the ability to handle the uncertainty caused by forecasted values, as the internal prices and the net power profile can be regulated with the actual values.

During the iteration process, each PV prosumer calculates its cost function with the changing of internal prices. As shown in Fig. 12, the cost of each PV prosumer oscillates with the increase of iterations and finally converges to a constant value. The converging process is consistent with the change of internal prices. It is noted that in order to amplify the changing tendency of PV prosumers' cost, the value of cost in Fig. 12 is normalized into the interval of [0, 1].

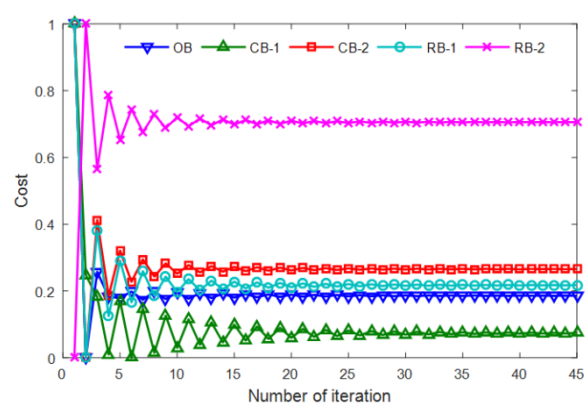

Fig. 12 Variation of PV prosumers' cost with the increase of iterations.

After all PV prosumers accept the internal prices and don't want to change the power consumptions, the final net power curves compared with original net power are shown in Fig. 11. From the results, the feed-in PV power to the utility grid

(mainly on 9:00 to 17:00) decreases obviously by adopting the proposed models.

The reasons for the aforementioned results (Figs. 9 and 12) can be concluded as follows. If the original SDR is large, the internal prices for selling and buying would be both very low. For selling role PV prosumers, they want to sell less power with the low price, while buying role PV prosumers may want to buy more power at this time. Consequently, both roles of PV prosumers would like to adjust the power consumption as to minimize the cost or maximize the profit. And the users' convenience level may also be affected, which reflects on their cost functions. Finally, the SDR is also reduced by the adjustment of load consumptions, which would enhance the value of internal prices. In turn, if the original SDR is small, the dynamical process will be inverted, and the SDR would be enhanced by the adjustment of load consumptions, which finally reduce the value of internal prices. According to the analysis in Section V, the internal prices pair of each time slot should finally be accepted by all PV prosumers, none of them would like to adjust the power consumption anymore when the internal price achieve the fixed point.

C. *Results of the cost and income*

The results of PV prosumers' costs and the ESP's income are listed in TABLE III.

TABLE III
VALUES OF PV PROSUMERS' COST AND ESP'S INCOME.

Objective	Feed-in-tariff (CNY)	Day-ahead optimization (CNY)	Hour-ahead optimization (CNY)
OB-1	1855.9	1721.6	1759.9
CB-1	2282.1	2201.0	2249.8
CB-2	4095.8	3900.5	3949.3
RB-1	1444.5	1364.4	1389.9
RB-2	2101.9	2002.0	2033.2
ESP	0	88.7	90.0
Total value	11780.1	11189.5	11382.1

The results show that the PV prosumers achieve effective cost savings compared with directly trading with the utility grid using feed-in-tariff. Whatever the roles of PV prosumers, it can benefit from the energy sharing model. As the agent, the ESP can also benefit from the price model to promote the PV energy sharing and its profit would be increased with the expanding number of PV prosumers. Moreover, the results show that cost saving of the hour-ahead optimization is lower than the day-ahead optimization. The reason is that the amount of sharing PV energy is less than expected due to the forecasting errors. This is a reasonable result that reflects the actual situation.

D. *Sensitive analysis of parameter α*

The parameter α is used to quantify users' willingness to the cost function and the DR penetration level. A larger α means that the user much more cares about the convenience degree and has a low level of DR penetration. With the change of α, the results of internal prices and economic results are compared in Fig. 13 and Table IV, respectively.

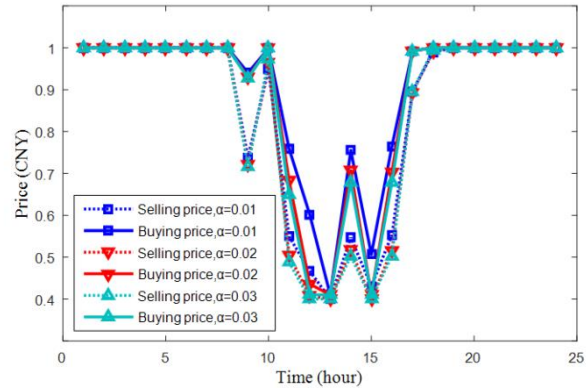

Fig. 13 Internal prices when α takes different values.

As shown in Fig. 13, the selling and buying prices reduce when α takes the value from 0.01 to 0.03, which means users' sensitivity to the price goes down with the enhancement of convenience level. Even if there is electricity with a lower price, they don't want to adjust the power consumption which may bother them. As shown in TABLE IV, PV prosumers' cost enhance with α's increasing except for CB-2. The reason is that CB-2's net power stays positive from 7:00 to 17:00, which means CB-2 always buys energy from other PV prosumers. When α increases, the buying price goes down and hence CB-2 can benefit from this situation. The ESP's income also decreases, which attributes to the less flexible power to promote the amount of sharing PV energy when α increases.

TABLE IV
COSTS OR INCOME WHEN α TAKES DIFFERENT VALUES.

Name	α = 0.01 (CNY)	α = 0.02 (CNY)	α = 0.03 (CNY)
OB-1	1710.5	1720.7	1726.4
CB-1	2193.6	2214.0	2223.5
CB-2	3885.9	3863.3	3861.2
RB-1	1360.7	1362.8	1368.4
RB-2	1975.8	2009.5	2021.3
ESP	77.0	76.3	75.8
Total value	11126.5	11170.3	11200.8

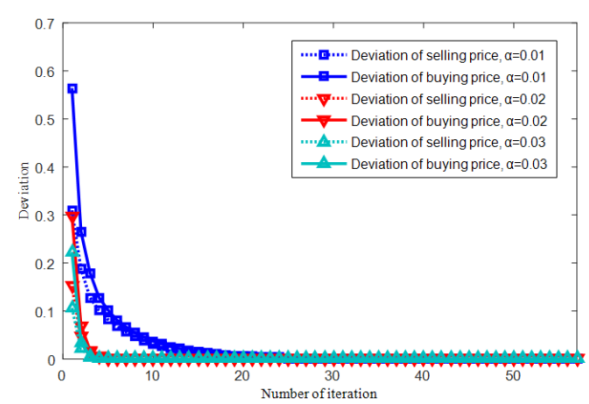

Fig. 14 Differences of selling price and buying price between two iterations when α takes different values.

The iteration processes of the energy sharing model with different α are shown in Fig. 14. As the DR penetration

would influence the convergence of optimization, the convergence becomes worse as α decreases from 0.03 to 0.01. As analyzed in [31] for DR problem, when users have more flexible power, factors that can influence the price become more, it's harder to get convergence on the internal price. In order to get roughly requirements for convergence, the result of convergence status with different α is shown in TABLE V.

TABLE V
CONVERGENCE TEST OF ALGORITHM WITH DIFFERENT α

α	0.015	0.012	0.010	0.009
Number of iterations	12	15	45	N/A
Convergence status	Converge	Converge	Converge	Diverge

The result shows that the setting of α would have great influence on the convergence. For each prosumer, the larger α, the more concern on the inconvenience would be caused by shifting loads, and thus it has less response to the price incentive. On the contrary, the smaller α_i makes prosumers to shift the load volatility, and then would greatly affect the convergence. From Table I, we can find the algorithm is diverged when $\alpha < 0.01$.

E. Analysis of technical effects and fairness

The technical effect on the distribution grid is an important issue. Several indexes of the effect before and after using the internal pricing are shown in TABLE VI. The results show that the proposed method has a positive effect on the distribution system with lower power loss, lower maximum load, and better Peak-to-average ratio. This is mainly due to that the locally consumed PV energy among prosumers are increased, and part of the loads are shifted with the SDR of PV energy.

TABLE VI
TECHNICAL EFFECT OF THE PROPOSED MODEL.

Index	Before using internal pricing	After using internal pricing
Power loss (kWh)	332.0	318.7
Maximum load (kW)	1152	1122.9
Peak-to-average	2.5214	2.4651

The result of fairness index *K* in the case day is 0.165. According to the definition of *K*, we cannot claim that the DR in energy sharing is absolutely fair. However, as compared with the DR allocation in the electricity market, this result is better than the normal load-based cost allocation method [33]. Having insight into this result, although the load shifting is a basic strategy to adjust the SDR and finally influence the internal prices, the costs and benefits of prosumers still have inner relations with their PV energy and willingness. These factors may have the impact on the fairness calculation.

F. Analysis of computation and data requirements

The results of convergence test for Algorithm 1 are listed in TABLE VII. The algorithm can approach the optimum solutions in less than 60 iterations (average 40 iterations). According to TABLE I, there are about 192 Bytes upstream data and 4.32 KB downstream data in one iteration for a UEMS. Thus, for one prosumer, the total data exchange for the day-ahead or hour-ahead optimization is less than 11.52 KB (upstream) and 259.2 KB (downstream). For the ESP, although the required data is increased with the number of prosumers, it still can be easily handled by normal computers. If we use wireless communication for data exchange, i.e., LTE 230MHz VPN network, which is recommended by the utility grid of China for related type of applications. The latency of each message is less than 3s (average 2s).

TABLE VII
CONVERGENCE TEST WITH DIFFERENT NUMBER OF PROSUMERS.

Number of Prosumers	5	10	15	20	25
Number of Iterations	59	55	39	49	43
Time cost (s)	240.7	224.4	159.0	199.9	175.4

Furthermore, we use a computer with Intel Core i5-4570 CPU 3.2GHz, 4G memory, and Matlab 2015b as the time cost testing environment for the algorithm. The average computation time is 0.0754s for solving the lower level objective function (23), combined with the communication latency, the total time cost of the day-ahead optimization is about 200 s. The time cost of hour-ahead optimization is even less than the day-ahead, since less data is required.

VI. CONCLUSION

In this paper, we have studied energy sharing inside a microgrid of P2P prosumers. We have formulated an internal price model and the cost model of prosumers considering the willingness of load shifting. The internal prices are co-decided by all the prosumers during the energy sharing. Moreover, the time horizon of the pricing model can be either day-ahead or hour-ahead. We have proposed a distributed iterative algorithm to solve the problem, which can be easily implemented with normal configuration for DR applications in the Smart Grid. Through the realistic data from a demonstration project, we have shown that by using the proposed model, the PV prosumers can achieve effective cost saving compared with directly trading with the utility grid under the feed-in-tariff, and the local consumption of PV energy can be increased as well.

The proposed work can be extended in various ways. An interesting extension would be constructing an energy sharing framework in distribution network level. Considering the electric distance among prosumers, the power loss during the energy sharing is an important factor [34] that should be integrated into the model. Furthermore, for the microgrid with islanded capability, the energy sharing can also be different. The internal pricing model could be designed according to the marginal cost of controllable distributed energy resources and battery energy storage system.

APPENDIX I

According to the basic principle of economics, the relation between price and SDR is inverse-proportional. Thus, the price decrease with the higher SDR and increased with the lower SDR. When SDR>1, there must be extra PV energy export to the utility grid, then the selling price could be set as λ^{sell} due to the constraints of feed-in tariff. When 0⩽SDR⩽1, as the sharing energy is lower than the load demand of the microgrid, the internal price can be set in accordance with the function of inverse proportion $1/(ax+b)$. Thus, we can use $1/(ax+b)$ as the fitting function to obtain the relationship between internal selling price and SDR:

$$Pr_{sell}^{h} = f(SDR) = \begin{cases} \frac{1}{aSDR+b}, & 0 \le SDR \le 1 \\ \lambda^{sell}, & SDR > 1 \end{cases} \tag{27}$$

If the SDR=0, there is no PV energy that can be shared in the energy sharing zone, that is, all the energy should be purchased from the utility grid with λ^{buy}. Thus, the function $f(SDR)$ should include the point $(0, \lambda^{buy})$. Otherwise, if SDR>1, the selling price is λ^{sell}, and thus the $f(SDR)$ should also include the point $(1, \lambda^{sell})$. By substituting the two points into the function $f(SDR)$, we can further obtain the following equation set:

$$\begin{cases} \frac{1}{a\times 0+b} = \lambda^{buy} \\ \frac{1}{a\times 1+b} = \lambda^{sell} \end{cases} \tag{28}$$

Solving the equation set, we can obtain:

$$\begin{cases} a = \frac{\lambda^{buy}-\lambda^{sell}}{\lambda^{buy}\lambda^{sell}} \\ b = \frac{1}{\lambda^{buy}} \end{cases} \tag{29}$$

Finally, by substituting (29) into (28), the function of internal selling price can be formulated as:

$$Pr_{sell}^{h} = f(SDR^{h}) = \begin{cases} \frac{\lambda^{sell}.\lambda^{buy}}{(\lambda^{buy}-\lambda^{sell})\cdot SDR^{h}+\lambda^{sell}}, & 0 \le SDR^{h} \le 1 \\ \lambda^{sell}, & SDR^{h} > 1 \end{cases} \tag{30}$$

Considering the economical balance requirement of the energy sharing zone, we can further deduce the internal buying price Pr_{buy}^{h} based on the internal selling price Pr_{sell}^{h}.

If $0 \le SDR \le 1$, since the ESP need to purchase energy from the utility grid, the economic balance can be described as:

$$TBP \cdot Pr_{buy} = TSP \cdot Pr_{sell} + (TBP - TSP)\lambda^{buy} \tag{31}$$

By substituting (7) into (31), there is:

$$TBP \cdot Pr_{buy} = TBP \cdot SDR \cdot Pr_{sell} + (TBP - TBP \cdot SDR)\lambda^{buy} \tag{32}$$

Thus, we can deduce the Pr_{buy} as:

$$Pr_{buy} = SDR \cdot Pr_{sell} + (1 - SDR)\lambda^{buy} \tag{33}$$

Otherwise, if $SDR > 1$, similar with the condition of internal selling price, the internal buying price is:

$$Pr_{buy} = \lambda^{sell} \tag{34}$$

Finally, the function of internal buying price can be formulated as:

$$Pr_{buy}^{h} = f(SDR^{h}) = \begin{cases} (Pr_{sell}^{h})SDR^{h} + \lambda^{buy}(1 - SDR^{h}), & 0 \le SDR^{h} \le 1 \\ \lambda^{sell}, & SDR > 1 \end{cases} \tag{35}$$

APPENDIX II

All PV prosumers participating in the DR are supposed to have a certain proportion of flexible load, which means they have the room for adjusting the size and time arrangement of the load. In this study, PV prosumers' load is comprised of two types, i.e., fixed load and shiftable load. Each prosumer is assumed to have a certain number of shiftable appliances or equipment. Such appliances or equipments may include PHEVs, washers, dryers, dishwashers, etc. [25]. The original power consumption of PV prosumer *i* at time slot *h* can be defined as:

$$\mathrm{TP}_{i}^{h} = \mathrm{sl}_{i0}^{h} + \mathrm{fl}_{i0}^{h} \tag{36}$$

where sl_{i0}^{h} is the original shiftable load of PV prosumer *i* at time slot *h*, and fl_{i0}^{h} is the original fixed load of PV prosumer *i* at time slot *h*.

As the fixed load has no contribution to the DR, adjusted power of the PV prosumer *i* at time slot *h* can be described as:

$$x_{i}^{h} = \mathrm{sl}_{i}^{h} + \mathrm{fl}_{i0}^{h} \tag{37}$$

where sl_{i}^{h} is the shiftable load of PV prosumer *i* at time slot *h*.

Assuming the number of shiftable appliances of prosumer *i* is K_i, the shiftable load of prosumer *i* at period *h* is the sum of all its shiftable appliances:

$$\mathrm{sl}_{i}^{h} = \sum_{k=1}^{K_i} \mathrm{sl}_{ik}^{h} \tag{38}$$

where sl_{ik}^{h} is the *k*-th shiftable appliance's adjusted value for prosumer *i* at time slot *h*. The detailed model of shiftable load involves the size, total amount, duration time and optional time range of the load:

$$\begin{cases} \mathrm{sl}_{ik}^{h} \in [\mathrm{sl}_{ik}^{\min}, \mathrm{sl}_{ik}^{\max}], & \mathrm{h} \in [\mathrm{t}_{ik}, \mathrm{t}_{ik} + \Delta\mathrm{T}_{ik}] \\ \mathrm{sl}_{ik}^{h} = 0, & \mathrm{h} \notin [\mathrm{t}_{ik}, \mathrm{t}_{ik} + \Delta\mathrm{T}_{ik}] \end{cases} \tag{39}$$

$$[\mathrm{t}_{ik}, \mathrm{t}_{ik} + \Delta\mathrm{T}_{ik}] \in [\alpha_{ik}, \beta_{ik}] \tag{40}$$

$$\sum_{h=\mathrm{t}_{ik}}^{\mathrm{t}_{ik}+\Delta\mathrm{T}_{ik}} \mathrm{sl}_{ik}^{h} = \mathrm{sq}_{ik} \tag{41}$$

In these equations, $[\mathrm{sl}_{ik}^{\min}, \mathrm{sl}_{ik}^{\max}]$ is the range of the *k*-th shiftable appliance for prosumer *i*, t_{ik} is the start time of the *k*-th shiftable appliance, $\Delta\mathrm{T}_{ik}$ is the duration time, $[\alpha_{ik}, \beta_{ik}]$ is the optional time range interval, and sq_{ik} is the energy demand amount of the *k*-th shiftable appliance for prosumer *i*. In addition, t_{ik} and $\mathrm{t}_{ik} + \Delta\mathrm{T}_{ik}$ need to be within the time range $[\alpha_{ik}, \beta_{ik}]$. For the sake of simplicity, it's assumed that $\mathrm{sl}_{ik}^{\min} = \mathrm{sl}_{ik}^{\max} = \mathrm{sl}_{ik}$ in (3), then sl_{ik}^{h} can be expressed as:

$$\mathrm{sl}_{ik}^{h} = \mathrm{sl}_{ik} \cdot \mathrm{sign}\{\max[(h - \mathrm{t}_{ik} + 1)(\mathrm{t}_{ik} + \Delta\mathrm{T}_{ik} - \mathrm{h})], 0\} \tag{42}$$

Similarly, the original value of the *k*-th shiftable load for prosumer *i* at time slot *h* can be expressed as:

$$\mathrm{sl}_{ik0}^{h} = \mathrm{sl}_{ik} \cdot \mathrm{sign}\{\max[(h - \mathrm{t}_{ik0} + 1)(\mathrm{t}_{ik0} + \Delta\mathrm{T}_{ik} - \mathrm{h})], 0\} \tag{43}$$

where t_{ik0} is the original start time of the *k*-th shiftable load.

The deviations between t_{ik} and t_{ik0}, sl_{ik}^{h} and sl_{ik0}^{h} can be denoted as δ_{ik} and $\Delta\mathrm{sl}_{ik}^{h}$, respectively:

$$\delta_{ik} = \mathrm{t}_{ik} - \mathrm{t}_{ik0} \tag{44}$$

$$\Delta\mathrm{sl}_{ik}^{h} = \mathrm{sl}_{ik}^{h} - \mathrm{sl}_{ik0}^{h} \tag{45}$$

For each shiftable load, the rearrangement of start time will diretly change the power consumption of time slot *h* by shift in and shift out. Thus, we can analyze the changes of $\Delta\mathrm{sl}_{ik}^{h}$, $h \in [1,2,\ldots,H]$ with δ_{ik}, as shown in TABLE VIII.

TABLE VIII
THE RELATIONSHIP BETWEEN SHIFTING TIME AND ENERGY

The range of δ_{ik}	Changing intervals	$\Delta\mathrm{sl}_{ik}^{h}$
$(-\infty, -\Delta\mathrm{T}_{ik})$	$\mathrm{h} \in [\mathrm{t}_{ik0} + \delta_{ik}, \mathrm{t}_{ik0} + \Delta\mathrm{T}_{ik} - 1 + \delta_{ik}]$	$+\mathrm{sl}_{ik}$
	$\mathrm{h} \in [\mathrm{t}_{ik0}, \mathrm{t}_{ik0} + \Delta\mathrm{T}_{ik} - 1]$	$-\mathrm{sl}_{ik}$
$[-\Delta\mathrm{T}_{ik}, -1]$	$\mathrm{h} \in [\mathrm{t}_{ik0} + \delta_{ik}, \mathrm{t}_{ik0} - 1]$	$+\mathrm{sl}_{ik}$
	$\mathrm{h} \in [\mathrm{t}_{ik0} + \Delta\mathrm{T}_{ik} + \delta_{ik}, \mathrm{t}_{ik0} + \Delta\mathrm{T}_{ik} - 1]$	$-\mathrm{sl}_{ik}$
$[1, \Delta\mathrm{T}_{ik}]$	$\mathrm{h} \in [\mathrm{t}_{ik0}, \mathrm{t}_{ik0} + \delta_{ik}]$	$-\mathrm{sl}_{ik}$
	$\mathrm{h} \in [\mathrm{t}_{ik0} + \Delta\mathrm{T}_{ik}, \mathrm{t}_{ik0} + \Delta\mathrm{T}_{ik} - 1 + \delta_{ik}]$	$+\mathrm{sl}_{ik}$
$(\Delta\mathrm{T}_{ik}, +\infty)$	$\mathrm{h} \in [\mathrm{t}_{ik0}, \mathrm{t}_{ik0} + \Delta\mathrm{T}_{ik} - 1]$	$-\mathrm{sl}_{ik}$
	$\mathrm{h} \in [\mathrm{t}_{ik0} + \delta_{ik}, \mathrm{t}_{ik0} + \Delta\mathrm{T}_{ik} - 1 + \delta_{ik}]$	$+\mathrm{sl}_{ik}$

According to TABLE VIII, we can conclude two conditions:

(1) If $\delta_{ik} > \Delta T_{ik}$, the total affected time slots is $2 \cdot \Delta T_{ik}$;

(2) If $\delta_{ik} \leq \Delta T_{ik}$, the total affected time slots is $2 \cdot \delta_{ik}$.

It means that the greater the δ_{ik}, the higher the changing probability of the *k*-th shiftable load at time slot *h*. Clearly, the more the shiftable appliances involved, the more amount the power consumption of prosumer *i* changes.

In rational terms, the inconvenience of power consumption could be measured by the number and time changes of the shiftable appliances. That is, we can use the changing degree of power consumption in the time horizon *H* to represent the inconvenience, as shown in (13). Moreover, in order to describe the different willingness of prosumers towards the shifting [24], we denote α_i as the sensitivity coefficient.

Finally, in order to simplify the subsequent calculation in this paper, we simplify the model of load shifting, which is regarded as a continuous variable, involving the size, total amount of the load:

$$\sum_{h=1}^{H} \mathrm{sl}_i^h = \sum_{h=1}^{H} \mathrm{sl}_{i_0}^h \tag{46}$$

$$\min(\mathbf{sl}_{i_0}) \leq \mathrm{sl}_i^h \leq \max(\mathbf{sl}_{i_0}) \tag{47}$$

These two expressions are equivalent to (17) and (18).

APPENDIX III

The DR actions of prosumers are related to the internal price $\boldsymbol{Pr}$. If the cost function of PV prosumers is convex, the optimal power consumption $\mathbf{x}_i$, $i \in [1,2,\dots,n]$ can be determined.

From equation (16), we can know that Pr_i^h is decided by the net power $x_i^h - P_i^h$. To prove the cost function of PV prosumer is strictly convex, three different situations are taken into consideration.

1) For each random x_i^{ha} and x_i^{hb}, $P_i^h \leq x_i^{ha} < x_i^{hb}$ or $x_i^{ha} < x_i^{hb} \leq P_i^h$, $0 < \theta < 1$. Then $Pr_i^h = Pr_{buy}^h$ or $Pr_i^h = Pr_{sell}^h$, respectively.

$$C_i^h\left(\theta \cdot x_i^{ha} + (1-\theta) \cdot x_i^{hb}\right) - \left[\theta \cdot C_i^h\left(x_i^{ha}\right) + (1-\theta) \cdot C_i^h\left(x_i^{hb}\right)\right] = -\alpha_i \cdot \theta \cdot (1-\theta) \cdot \left(x_i^{ha} - x_i^{hb}\right)^2 \tag{48}$$

As $\alpha_i > 0$, so

$$C_i^h\left(\theta \cdot x_i^{ha} + (1-\theta) \cdot x_i^{hb}\right) < \theta \cdot C_i^h\left(x_i^{ha}\right) + (1-\theta) \cdot C_i^h\left(x_i^{hb}\right) \tag{49}$$

2) For each random x_i^{ha} and x_i^{hb}, $x_i^{ha} < P_i^h < x_i^{hb}$, $0 < \theta < 1$ and $x_i^{ha} < \theta \cdot x_i^{ha} + (1-\theta) \cdot x_i^{hb} < P_i^h < x_i^{hb}$. Then $Pr_i^h\left(x_i^{ha}\right) = Pr_{sell}^h, Pr_i^h\left(\theta \cdot x_i^{ha} + (1-\theta) \cdot x_i^{hb}\right) = Pr_{sell}^h$, and $Pr_i^h(x_i^{hb}) = Pr_{buy}^h$, respectively.

$$C_i^h\left(\theta \cdot x_i^{ha} + (1-\theta) \cdot x_i^{hb}\right) - \left[\theta \cdot C_i^h\left(x_i^{ha}\right) + (1-\theta) \cdot C_i^h\left(x_i^{hb}\right)\right] = -\alpha_i \cdot \theta \cdot (1-\theta) \cdot \left(x_i^{ha} - x_i^{hb}\right)^2 + (1-\theta) \cdot (Pr_{buy}^h - Pr_{sell}^h) \cdot (P_i^h - x_i^{hb}) \tag{50}$$

As $\alpha_i > 0, Pr_{buy}^h \geq Pr_{sell}^h$, so

$$C_i^h\left(\theta \cdot x_i^{ha} + (1-\theta) \cdot x_i^{hb}\right) < \theta \cdot C_i^h\left(x_i^{ha}\right) + (1-\theta) \cdot C_i^h\left(x_i^{hb}\right) \tag{51}$$

3) For each random x_i^{ha} and x_i^{hb}, $x_i^{ha} < P_i^h < x_i^{hb}$, $0 < \theta < 1$ and $x_i^{ha} < P_i^h \leq \theta \cdot x_i^{ha} + (1-\theta) \cdot x_i^{hb} < x_i^{hb}$. Then $Pr_i^h\left(x_i^{ha}\right) = Pr_{sell}^h, Pr_i^h\left(\theta \cdot x_i^{ha} + (1-\theta) \cdot x_i^{hb}\right) = Pr_{buy}^h$, and $Pr_i^h(x_i^{hb}) = Pr_{buy}^h$, respectively.

$$C_i^h\left(\theta \cdot x_i^{ha} + (1-\theta) \cdot x_i^{hb}\right) - \left[\theta \cdot C_i^h\left(x_i^{ha}\right) + (1-\theta) \cdot C_i^h\left(x_i^{hb}\right)\right] = -\alpha_i \cdot \theta \cdot (1-\theta) \cdot \left(x_i^{ha} - x_i^{hb}\right)^2 + \theta \cdot (Pr_{buy}^h - Pr_{sell}^h) \cdot (x_i^{ha} - P_i^h) \tag{52}$$

As $\alpha_i > 0, Pr_{buy}^h \geq Pr_{sell}^h$, so

$$C_i^h\left(\theta \cdot x_i^{ha} + (1-\theta) \cdot x_i^{hb}\right) < \theta \cdot C_i^h\left(x_i^{ha}\right) + (1-\theta) \cdot C_i^h\left(x_i^{hb}\right) \tag{53}$$

Thus, $C_i^h\left(x_i^h\right)$ is a strictly convex function of x_i^h, and C_i is a strictly convex function on $\boldsymbol{x}$ ($\boldsymbol{x} = [x_i^1, x_i^2, \dots, x_i^H]$), so the optimal scheduling of $\boldsymbol{x}$ is one-to-one correspondence to the internal price $\boldsymbol{Pr}$. In other words, $\mathbf{x}$ is the function of $\mathbf{Pr}$: $\mathbf{x} = Y(\mathbf{Pr})$. In addition, according to the internal pricing model (9)-(10), the relationship between $\boldsymbol{Pr}$ and $\boldsymbol{x}$ can be described as $\boldsymbol{Pr} = H(\boldsymbol{x})$. So the whole process can be described as:

$$\boldsymbol{Pr} = H\left(Y(\mathbf{Pr})\right) = V(\mathbf{Pr}) \tag{54}$$

where *V* is used to indicate the relationship of $\mathbf{Pr}$. As *H* and *Y* are compact and continues, function V is also compact and continues.

Theorem (*Brouwer Fixed Point Theorem*) Let X be a subset of R^N, and $f: X \to R^N$. When X is bounded, closed and convex set; *f* is compact, continues and $f: X \to X$. Then *f* has a fixed point X [30].

It is easy to figure out that the internal price pair $\mathbf{Pr} \in D$, and *D* is a bounded closed convex set.

$$D = \{(Pr_{buy}^1, Pr_{buy}^2, \dots, Pr_{buy}^H; Pr_{sell}^1, Pr_{sell}^2, \dots, Pr_{sell}^H) | Pr_{buy}^h \in [\lambda^{sell}, \lambda^{buy}],\ Pr_{sell}^h \in [\lambda^{sell}, \lambda^{buy}],\ h = 1,2,\dots,H\}.$$

As $\mathbf{Pr} \in D$ and $V: D \to D$, so there exists a fix point in *D* for $V(\mathbf{Pr})$ according to the **Theorem**.

APPENDIX IV

From the proposed model and algorithm, the internal prices are finally co-decided by all the participant prosumers on an iterative way. During the process, prosumers may adjust different quantities of shiftable load and gain different benefits of DR. Thus, it is important to guarantee the fairness of the energy sharing. There are several indexes are proposed, i.e., [33] [35], etc. Considering the similarity of application scenarios, we have chosen the fairness definition in [33] as the basic principle, which is defined as the variance in the benefit to cost ratios of market participants located at each node. In the context of energy sharing zone, a fairness index *K* is defined as variance in the economic benefit to load shifting quantity of participant prosumers, which is expressed as follows.

$$K = \frac{1}{n}\sum_i^n \left(\frac{EB_i}{Sh_i} - \frac{1}{n}\sum_{i=1}^n \frac{EB_i}{Sh_i}\right)^2 \tag{55}$$

$$EB_i = C_i^e - C_i^{e\prime} \tag{56}$$

$$Sh_i = \frac{\sum_{h=1}^{H}\left|TP_i^h - x_i^h\right|}{2} \tag{57}$$

where EB_i is the economic benefit of prosumer i, C_i^e and $C_i^{e\prime}$ is energy consumption cost of prosumer *i*, by measuring as before and after participating in energy sharing zone, respectively. Sh_i is total energy of load shifting (kWh) of prosumer *i* in 24h.

If all participants are shifted loads exactly proportionally to their economic benefits, then K=0. Large values of K indicate that some prosumers have been allocated more or less than

their “fair share” of contributions. After every day’s operation, the ESP can publish the result of fairness index to every prosumer. This result may affect the participating willingness of prosumers.

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